

S4 File Supporting Information: Code Workflow

1. Exposure Factor Analysis
 - a. Script: S1_script_exposure_analysisMS.R
 - b. Inputs
 - i. NetCDF (.nc) files of exposure factors
 - ii. Finalized shapefiles of species distributions
 - c. Runs full exposure analysis for each species for the selected exposure factors
 - d. Outputs
 - i. Maps of each exposure factor over each species distribution
 - ii. Plots of histograms and summarized bar plots of anomalies of each exposure factor for each species distribution categorized into low, moderate, high, and very high bins based on anomaly value
 - iii. Table of exposure score breakdown for each species, including the weighted mean score for each exposure factor
 - iv. Table of final exposure factor scores across species (low, moderate, high, or very high)
 - v. A figure of the distribution of mean scores for each exposure factor across species (Fig. 4a)
2. Sensitivity Attribute Analysis
 - a. Script: S2_script_sensitivity_attr_scoringMS.R
 - b. Inputs
 - i. Sensitivity attribute scores from the scientific panelists as a result of the scoring workshop (S1 Dataset: Biological Sensitivity Attribute Scores)
 - c. Runs full sensitivity attribute analysis for each species
 - d. Runs distribution shift potential analysis for each species (using a subset of four sensitivity attributes)
 - e. Outputs
 - i. PDF of stacked barplots of scientific panelist scores for each sensitivity attribute for each species from the workshop, color-coded by panelist
 1. And combined .pdfs for each species group
 - ii. Table of sensitivity attribute score breakdown for each species, including the weighted mean score (and standard deviation) for each sensitivity attribute and data quality mean
 1. Separate table with all species score breakdowns in one table
 - iii. Table of final sensitivity attribute scores across species (low, moderate, high, or very high)
 - iv. A figure of the distribution of mean scores for each sensitivity attribute across species (Fig. 4b)
 - v. Table of distribution shift potential ranks across species (low, moderate, high, or very high)
3. Data Quality
 - a. Script: S3_script_data_quality_scoresMS.R
 - b. Inputs
 - i. Sensitivity attribute scores and accompanying data quality scores from the scientific panelists as a result of the scoring workshop (S1 Dataset: Biological Sensitivity Attribute Scores)

- ii. List of species names along with other names and abbreviations used throughout scripts (S2 Table: Species Name Match List)
 - c. Calculates mean and standard deviation of data quality scores for each species
 - d. Calculate mean and standard deviation of data quality scores for each attribute
 - e. Outputs
 - i. Table of mean and standard deviation of data quality scores across species
 - ii. Barplots by species group where each bar represents the mean and standard deviation of the data quality scores for each species (Fig. 10)
 - iii. Barplots by species group where each bar represents the mean and standard deviation of the data quality scores for each attribute (Fig. 11)
4. Vulnerability Rankings
 - a. Script: S4_script_vulnerability_rankingMS.R
 - b. Inputs
 - i. Table of final exposure factor scores across species (low, moderate, high, or very high) (exposure scores generated from S1 Script)
 - ii. Table of final sensitivity attribute scores across species (low, moderate, high, or very high) (sensitivity attribute scores generated from S2 Script)
 - c. Calculates final vulnerability ranking for each species from the exposure factor score and sensitivity attribute score
 - d. Outputs
 - i. Table with exposure and sensitivity scores and final vulnerability ranking for each species
5. Vulnerability ranking uncertainty, distribution potential uncertainty, directional effects, directional effects uncertainty, leave-one-out analysis
 - a. Script: S5_script_vulnerability_uncertaintyMS.R
 - b. Inputs
 - i. Sensitivity attribute scores from the scientific panelists as a result of the scoring workshop (S1 Dataset: Biological Sensitivity Attribute Scores)
 - ii. Table of final exposure factor scores across species (low, moderate, high, or very high) (exposure scores generated from S1 Script)
 - iii. Table of final sensitivity attribute scores across species (low, moderate, high, or very high) (sensitivity attribute scores generated from S2 Script)
 - iv. Table that has various names of each species since throughout process the common name, shortened common name, scientific name, and abbreviations are used (S2 Table: Species Name Match List)
 - v. Table with exposure and sensitivity scores and final vulnerability ranking for each species (vulnerability scores generated from S4 Script)
 - vi. Table with abbreviations for each exposure factor (S3 Table: Exposure Factor Abbreviations)
 - vii. Table of distribution shift potential ranks across species (low, moderate, high, or very high) (distribution shift potential scores generated from S2 Script)
 - viii. Table of directional effects from the scientific panelists as a result of the scoring workshop (S2 Dataset: Directional Effect Scores)
 - c. Calculates uncertainty around the vulnerability ranking and distribution shift potential by resampling the scientific panelists' scores and bootstrapping 10,000 times per species
 - d. Calculate directional effects mean and ranks for each species and uncertainty by resampling the scientific panelists' directional effects scores and bootstrapping 10,000 times per species

- e. Leave-One-Out Analysis to assess sensitivity of vulnerability ranking to each sensitivity attribute and exposure factor
- f. Outputs
 - i. Table of uncertainty percent for vulnerability and distribution shift potential for each category (low, moderate, high, very high) for each species
 - ii. Barplot of number of species that are in each category (low, moderate, high, very high) for distribution shift potential
 - iii. Table of uncertainty percent for directional effects for each category (negative, neutral, positive) for each species
 - iv. Barplot of number of species that are in each category (negative, neutral, positive) for directional effects
 - v. Table with exposure and sensitivity scores, final vulnerability ranking, final vulnerability rank uncertainty, distribution shift potential score, distribution shift potential score uncertainty, directional effects mean, directional effects rank, and directional effects uncertainty for each species
 - vi. Figure of number of species that changed vulnerability ranking when each sensitivity attribute was left out (Fig. 5b)
 - vii. Figure of number of species that changed vulnerability ranking when each exposure factor was left out (Fig. 5a)
 - viii. Figure of a series of plots showing number of species with overall vulnerability ranking (low, moderate, high, very high) by species group, number of species with distribution shift potential ranking (low, moderate, high, very high) by species group, and the number of species with directional effects ranking (negative, neutral, positive) by species group (Fig. 6)
- 6. Non-metric multidimensional scaling analysis
 - a. Script: S6_script_nMDS_analysisMS.R
 - b. Inputs
 - i. Table with all species score breakdowns in one table (detailed sensitivity attribute scores generated in S2 Script)
 - ii. Table with exposure and sensitivity scores, final vulnerability ranking, final vulnerability rank uncertainty, distribution shift potential score, distribution shift potential score uncertainty, directional effects mean, directional effects rank, and directional effects uncertainty for each species (final vulnerability rankings with all other scores generated in S5 Script)
 - c. Run nMDS analysis to see if sensitivity attribute means are shared by species group and ANOSIM to see if sensitivity attribute means are statistically more similar within species group
 - d. Run nMDS analysis to see if sensitivity attribute means are shared by species with the same vulnerability rank and ANOSIM to see if sensitivity attribute means are statistically more similar for species with the same vulnerability rank
 - e. Outputs
 - i. Figure of nMDS of sensitivity attribute by species group (Fig. 7a)
 - ii. Figure of nMDS of sensitivity attribute by vulnerability rank (Fig. 7b)
- 7. Species Narratives
 - a. Script: S8_script_species_narrativesMS.R and script with image making function (S7_script_species_narratives_funcMS.R)
 - b. Inputs

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- i. Table that has various names of each species since throughout process the common name, shortened common name, scientific name, and abbreviations are used (S2 Table: Species Name Match List)
 - ii. Table with abbreviations for each exposure factor (S3 Table: Exposure Factor Abbreviations)
 - iii. Table with exposure and sensitivity scores, final vulnerability ranking, final vulnerability rank uncertainty, distribution shift potential score, distribution shift potential score uncertainty, directional effects mean, directional effects rank, and directional effects uncertainty for each species (final vulnerability rankings with all other scores generated in S5 Script)
 - iv. Table of mean and standard deviation of data quality scores across species (sensitivity attribute data quality scores generated in S3 Script)
- c. Create species narrative image for each species
- d. Outputs
 - i. Species narrative image for each species